CS2 Week 2

Diagonalization, Modal Analysis, Reachability and Observability

Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Short quizes and tips for PS
- Notebook available for some examples

Notebook on Github

Haixiao's Polybox

Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback Pole Placement, LQR	3
State Estimation, Luenberger Obs., Kalman Filter	4
03/17 Separation Principle, LQG, Stability margins	5
Intro to MIMO Systems, Transmission Zeros	6
Nonlinear Systems	7
break	8
Discrete-time Unconstrained Optimal Control	9
Convex Optimization	10
Model Predictive Control	11
Practical Model Predictive Control	12
Robust Model Predictive Control	13
Implementation Methods	14

0. Recap Week 1 in Notebook

Today

1. **Diagonalization**
2. **Modal Analysis**
3. **Reachability/ Observability**
4. **Kalman Decomposition**
5. **Popov-Belevitch-Hautus Test**

1. Diagonalization

Similarity Transformation

$$\tilde{x} = T^{-1}x$$

$$\tilde{A} = T^{-1}AT$$

$$\tilde{B} = T^{-1}B$$

$$\tilde{C} = CT$$

1. **Remember from LinAlg, transform system A with invertible matrix T**

2. **Using proof, we notice the TF and time response stay the same after transformation**

$$\begin{aligned}\tilde{G}(s) &= \tilde{C}(sI - \tilde{A})^{-1}\tilde{B} + \tilde{D} = CT(sI - T^{-1}AT)^{-1}T^{-1}B \\ &= C(T(sI - T^{-1}AT)T^{-1})^{-1}B + D = C(sI - A)^{-1}B + D \\ &= G(s).\end{aligned}$$

3. **We notice, that a state space representation of a system is NOT UNIQUE**

4. **Soooo, what is the most effective representation?**

Diagonalization!

1. **If matrix A is diagonalizable, there exists a matrix** $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$

2. **And A can be represented in the following way where V is assembled from the Eigenvectors of A** $\rightarrow$ $A = V \Lambda V^{-1}$

3. **Many advantages to have a system diagonalized**
   - **Easy calculation of exponential** $\rightarrow$ $e^{At} = V e^{\Lambda t} V^{-1} \longrightarrow e^{\Lambda t} = \mathrm{diag}(e^{\lambda_1 t}, \ldots)$
   - **Detect stability**
   - **Decoupled Modes**

Quiz 1

What does it mean if the Eigenvalue has an imaginary component?

A. The system is automatically stable

B. The system has a pole at 0

C. The system shows oscillatory behavior

D. None of the above

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2. Modal Analysis

(Way to determine system behavior)

Modal decomposition

1. **The homogeneous solution of a system can be found with** $\tilde{x}(t) = e^{\tilde{A}t}\tilde{x}(0)$
   - **Hard to calculate!**

2. **So we use** $\tilde{A} = \Lambda$ **with diagonalization, we can also say** $\tilde{x}_i(t) = e^{\lambda_i t}\tilde{x}_i(0).$

3. **We then use the** $x(0) = V\tilde{x}(0)$ **, remember** $\tilde{x} = T^{-1}x$, $\tilde{A} = T^{-1}AT$

$$x(t) = e^{At}x(0) = e^{At}V\tilde{x}(0) = Ve^{\Lambda t}\tilde{x}(0) = \sum_{i=1}^{n} e^{\lambda_i t}\tilde{x}_i(0)\, v_i.$$

$$e^{At} = Ve^{\Lambda t}V^{-1}$$

Modal analysis

1. **Each of the value in the i in** $x(t) = \sum_{i=1}^{n} e^{\lambda_i t} \tilde{x}_i(0)\, v_i.$ **represents a mode of system x**

2. **The behavior in each of these modes can be determined seeing the eigenvalue:**

   - **Large absolute real part->** **quick response**
   - **Negative real part ->** **system goes to zero**
   - **Existing imaginary part ->** **oscillations**
   - **Multiple same value ->** **polynomial scaling**

3. Reachability/ Observability

Reachability

1. **Definition**: Reachable, if there exists a signal, which can take a system with initially x = 0 to any arbitrary state x_final

2. Based on modal decomposition, we know a **mode i is reachable, if** $\tilde{b}_i \neq 0$

3. But that does not tell, whether the entire system in reachable, what if:
   - Each of the mode is reachable
   - But states are not independent!
   - In this case, **system not reachable!**

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

Reachability

1. **Better solution: We use the reachability matrix:** $R = [B \; AB \; A^2B \ldots A^{n-1}B]$

2. **A system is reachable, if** $\operatorname{rank}(R) = n$ **, where n is the dimension of A**

3. **Goes for CT and DT systems**

Observability

1. **Definition: Oberservable, if any arbitrary initial condition x can be reconstructed using only the input and output over a finite time**

2. **Similar to reachability, there exists the observability matrix** $$O = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

3. **A system is said to be observable, if** $\operatorname{rank}(O) = n$

4. **Also works in both DT and CT**

Quiz 2

Given this observability matrix, is the system observable?

$$O = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix}$$

A. True

B. False

Quiz 2

Given this observability matrix, is the system observable?

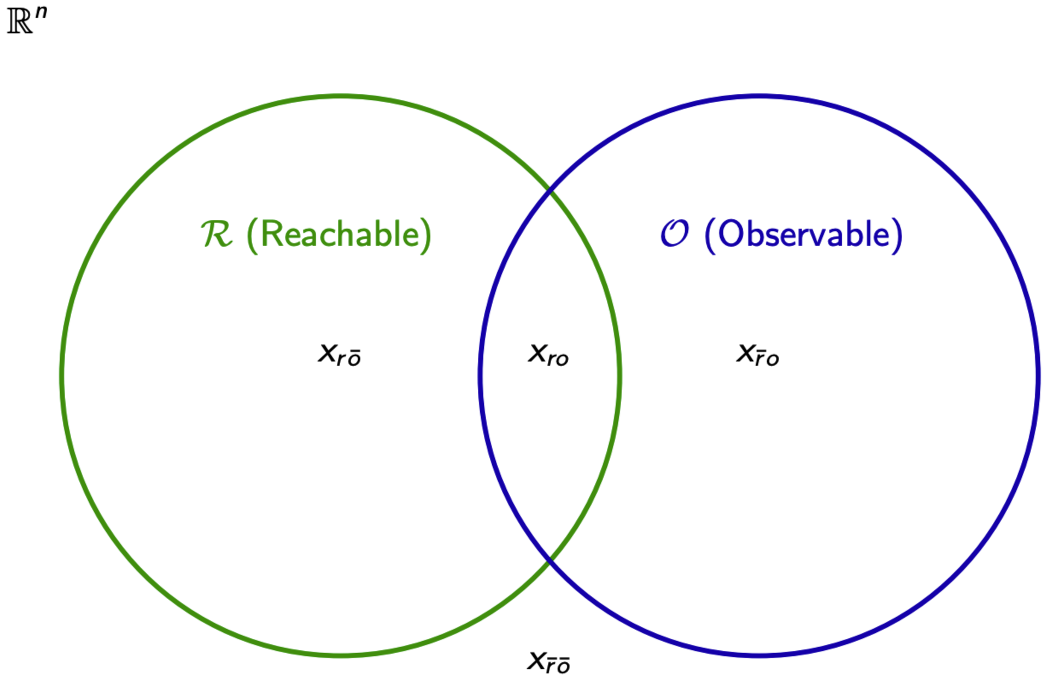
$$O = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix}$$

A. True

B. False

4. Kalman Decomposition

Reachable/Observable spaces



Kalman decomposition

1. **In a TF, only the states which are both reachable and observable matter**

2. **The other states exist, but not noticeable based on input and output**

3. **Kalman decomposition isolates each of the spaces**
   - **Reveals minimal realization**
   - **Shows hidden states not visible in TF**

$$
\begin{aligned}
x^{+} &= \begin{bmatrix} A_{r\bar{o}} & A_{12} & A_{13} & A_{14} \\ 0 & A_{ro} & 0 & A_{24} \\ 0 & 0 & A_{\bar{r}\bar{o}} & A_{34} \\ 0 & 0 & 0 & A_{\bar{r}o} \end{bmatrix} x + \begin{bmatrix} B_{r\bar{o}} \\ B_{ro} \\ 0 \\ 0 \end{bmatrix} u, \\
y &= \begin{bmatrix} 0 & C_{ro} & 0 & C_{\bar{r}o} \end{bmatrix} x + Du.
\end{aligned}
$$

Stabilizability/ Detectability

1. **If you are interested in how to perform a full Kalman Decomposition, see an example in:**

2. **Using the Kalman decomposed system, more system dynamics are revealed**
   - **A system is stabilizable, if all unstable modes are reachable**
   - **A system is detectable, if all unstable modes are observable**

Quiz 3

Which components need to have Re(λ) < 0 for a system to be stabilizable

$$
\begin{aligned}
x^+ &= \begin{bmatrix} A_{r\bar{o}} & A_{12} & A_{13} & A_{14} \\ 0 & A_{ro} & 0 & A_{24} \\ 0 & 0 & A_{\bar{r}\bar{o}} & A_{34} \\ 0 & 0 & 0 & A_{\bar{r}o} \end{bmatrix} x + \begin{bmatrix} B_{r\bar{o}} \\ B_{ro} \\ 0 \\ 0 \end{bmatrix} u, \\
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$A_{r\bar{o}}$	$A_{\bar{r}o}$	$A_{\bar{r}\bar{o}}$	$B_{r\bar{o}}$	B_{ro}

Quiz 3

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\end{aligned}
$$

$A_{r\bar{o}}$	$A_{\bar{r}o}$	$A_{\bar{r}\bar{o}}$	$B_{r\bar{o}}$	B_{ro}

5.Popov-Belevitch-Hautus Test

Popov-Belevitch-Hautus Test (PBH)

Method to determine stabilizability and detectability without a full Kalman decomp.

- PBH test for controllability (same as reachability for continuous time systems):

$$\operatorname{rank}\begin{bmatrix}\lambda I - A & B\end{bmatrix} = n \quad \forall\lambda \in \mathbb{C}.$$

- PBH test for stabilizability:

$$\operatorname{rank}\begin{bmatrix}\lambda I - A & B\end{bmatrix} = n \quad \forall\lambda \in \mathbb{C} \text{ with } \operatorname{Re}(\lambda) > 0.$$

- PBH test for observability:

$$\operatorname{rank}\begin{bmatrix}\lambda I - A \\ C\end{bmatrix} = n \quad \forall\lambda \in \mathbb{C}.$$

- PBH test for detectability:

$$\operatorname{rank}\begin{bmatrix}\lambda I - A \\ C\end{bmatrix} = n \quad \forall\lambda \in \mathbb{C} \text{ with } \operatorname{Re}(\lambda) > 0.$$

Summary of Today

1. **Diagonalization of systems** $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ $A = V\Lambda V^{-1}$

2. **Modal Analysis** $x(t) = \sum_{i=1}^{n} e^{\lambda_i t} \tilde{x}_i(0)\ v_i.$ **and system behavior**

3. **Reachability/Observability** $R = [B\ AB\ A^2B \ldots A^{n-1}B]$ $O = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$

4. **Stabilizability/Detectability using Kalman Decomp. or PBH-Test**

Tips for Problem Set 2

1: "Controllable" means that we can take any state back to 0, in CT, they are identical!
In DT, reachable implies controllable, but not the other way!

2: b) Use $x(t) = \sum_{i=1}^{n} e^{\lambda_i t} \tilde{x}_i(0) v_i, \qquad \tilde{x}(0) = T^{-1} x(0).$ **c) Plot modes as arrows**

3: Triangular Matrix? Use that property!

4: Write down the state-space matrices first! Plug in the factors

Thank you for listening!

Feedback?

Too fast? Too slow? Fewer theory, more exercises?

I would appreciate your feedback. Please let me know.